

Hoang Ngoc Loc

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SUMMARY

Backend engineer with 2+ years building production Node.js / TypeScript services for international clients — an 18-service HR-tech platform on Kubernetes and an AI compliance SaaS on Azure. Strong in event-driven architecture, PostgreSQL correctness under concurrency, and test-heavy delivery (Testcontainers, gated CI/CD, OpenTelemetry). Started full-stack, now focused fully on backend and distributed systems; comfortable owning services end-to-end and working in English with distributed teams.

WORK EXPERIENCE

Full-stack Developer — 24H.DEV CO., LTD

Jul 2024 – Jul 2026 | Da Nang, Vietnam

- Owned and shipped production features across 18+ Node.js microservices on a two-sided recruitment platform for an HR-tech client — BFF gateways, profile, jobs, messaging, CV-parser, matches-scheduler — deployed on GKE.
- Built an event-driven CDC pipeline that keeps matching and profile data consistent across services in near real time, replacing periodic batch sync.
- Rebuilt the candidate-search pipeline under production traffic: traced search/profile drift to slow queries and stale-read paths, then reshaped the indexing and write flow so matches stay correct as volume grows.
- Designed and shipped a FATCA/CRS compliance platform for a European fintech client: REST + GraphQL APIs on PostgreSQL, Azure AD SSO, and a LangGraph + Azure OpenAI classification pipeline with Document Intelligence — cutting filing-preparation time by 60–80%.
- Also delivered several smaller outsourced products end-to-end as sole developer in short engagements — Node.js / TypeScript backend, API design and data modelling, plus the React / Next.js front-end.

Front-end Developer — MLOps VN

Mar 2023 – Sep 2023 | Remote

- Built and shipped the official site for MLOps Marathon 2023, Vietnam's first MLOps competition — responsive Next.js + SCSS Modules on Vercel: mlops.vn.

PERSONAL PROJECTS

jCool E-Commerce Backend

2026

NestJS, TypeScript, PostgreSQL + Drizzle, Redis, BullMQ, Meilisearch, S3, Stripe, Docker

Code: github.com/jCool10/jcool-ecommerce-backend | Live: jcool-ecommerce.up.railway.app

- Modular monolith with 7 bounded contexts and 53 REST endpoints; Clean Architecture / DDD boundaries enforced in CI by dependency-cruiser so any context can be extracted into a service without a schema migration.
- Concurrency-safe checkout: stock reserved inside the checkout transaction (no oversell), idempotent Stripe payments (no double-charge), Transactional Outbox → BullMQ with at-least-once delivery and idempotent consumers, payment saga with compensation.
- 1,131 unit + 68 integration tests on real Postgres/Redis/S3/SMTP via Testcontainers (no DB mocking), gated CI/CD, OpenTelemetry tracing and Prometheus SLO alerts.

RAG Chat with PDF

2025

Express.js, TypeScript, Python FastAPI, LangChain, Milvus, MongoDB, AWS S3

Code: github.com/jCool10/RAG-chat-with-pdf

- Two-service RAG system for Q&A over PDFs — Node.js + Express API and a Python FastAPI retrieval service — hybrid search (dense + BM25) on Milvus with LangChain embeddings, JWT auth and S3 storage.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, Python
Backend & Architecture	Node.js, NestJS, Express.js, REST, GraphQL (Apollo), Microservices, Event-driven (Google Cloud Pub/Sub, CDC, BullMQ), Transactional Outbox, Saga, Idempotency, DDD / Clean Architecture
Data	PostgreSQL (indexing, query tuning, transactions & concurrency control), MongoDB, Redis, Drizzle ORM, Prisma, pgvector
Cloud & DevOps	GCP (GKE, Pub/Sub), Azure (AD, OpenAI, Document Intelligence, Blob), AWS S3, Docker, Kubernetes, GitHub Actions, CI/CD, Nginx
Testing & Observability	Vitest, Testcontainers, k6, OpenTelemetry, Prometheus, Grafana
AI / LLM	LangChain, LangGraph, OpenAI & Anthropic APIs, RAG, Milvus
Front-end	React.js, Next.js, Tailwind CSS

EDUCATION

Da Nang University of Science and Technology — Bachelor of Computer Engineering

Da Nang, Vietnam | 2020 – 2025